

IND320 project work part 1

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GitHub link: https://github.com/Minajohanne/IND320_project_Minajohanne

Streamlit link: <https://ind320projectminajohanne.streamlit.app/>

Recording link: <https://nmbu.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=9c1bb640-9aff-47b8-83c9-b4d300f146c1>

Import packages

```
In [1]: # data handling
import pandas as pd

# visualizations
import plotly.express as px
import plotly.io as pio
import matplotlib.pyplot as plt
```

Reading the data

```
In [2]: # reading in the data
dat = pd.read_csv('../data/reservoirs.csv')
dat.head()
```

```
Out[2]:
```

	dato_Id	omrType	omrnrr	iso_aar	iso_uke	fyllingsgrad	kapasitet_TWh	fylling
0	1995-09-03	EL	4	1995	35	0.944721	21.079208	19.9
1	2020-11-15	EL	1	2020	46	0.974361	6.003264	5.84
2	2011-08-28	EL	4	2011	34	0.786499	21.079208	16.5
3	1998-07-05	EL	3	1998	27	0.793969	8.920932	7.01
4	2011-03-27	EL	4	2011	12	0.300820	21.079208	6.3

Data exploration

```
In [3]: # dimensions
dat.shape
```

```
Out[3]: (14877, 11)
```

```
In [4]: dat.describe()
```

Out [4]:

	omrn	iso_aar	iso_uke	fyllingsgrad	kapasitet_TWh	
count	14877.000000	14877.000000	14877.000000	14877.000000	14877.000000	1
mean	2.333333	2010.346038	26.405929	0.618852	29.145860	
std	1.490762	9.146696	15.017910	0.214850	22.739070	
min	0.000000	1995.000000	1.000000	0.055160	6.003264	
25%	1.000000	2002.000000	13.000000	0.456234	17.390587	
50%	2.000000	2010.000000	26.000000	0.648049	23.237715	
75%	3.000000	2018.000000	39.000000	0.802135	34.043590	
max	5.000000	2026.000000	53.000000	1.020072	87.437580	

In [5]: `dat.info()`

```

<class 'pandas.DataFrame'>
RangeIndex: 14877 entries, 0 to 14876
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   dato_Id                               14877 non-null  str
1   omrType                               14877 non-null  str
2   omrn                                  14877 non-null  int64
3   iso_aar                              14877 non-null  int64
4   iso_uke                              14877 non-null  int64
5   fyllingsgrad                         14877 non-null  float64
6   kapasitet_TWh                       14877 non-null  float64
7   fylling_TWh                         14877 non-null  float64
8   neste_Publiseringdato               14877 non-null  str
9   fyllingsgrad_forrige_uke           14877 non-null  float64
10  endring_fyllingsgrad                14877 non-null  float64
dtypes: float64(5), int64(3), str(3)
memory usage: 1.7 MB

```

Renaming the column names to english and understandable names.

```

In [6]: # map names dictionary
map_names = {
    "dato_Id": "date",
    "omrType": "area_type",
    "omrn": "area_number",
    "iso_aar": "iso_year",
    "iso_uke": "iso_week",
    "fyllingsgrad": "filling_level",
    "kapasitet_TWh": "capacity_TWh",
    "fylling_TWh": "stored_energy_TWh",
    "neste_Publiseringdato": "next_publication_date",
    "fyllingsgrad_forrige_uke": "filling_level_previous_week",
    "endring_fyllingsgrad": "change_in_filling_level"
}

# map names
dat = dat.rename(columns=map_names)

```

Converting column "date" to a datetime object and sorting the dates.

```
In [7]: dat["date"].dtype
```

```
Out[7]: <StringDtype(na_value=nan)>
```

```
In [8]: # convert date to a datetime object
dat["date"] = pd.to_datetime(dat["date"])
dat["date"].dtype
```

```
Out[8]: dtype('<M8[us]>')
```

```
In [9]: dat = dat.sort_values("date")
dat.head()
```

```
Out[9]:
```

	date	area_type	area_number	iso_year	iso_week	filling_level	capacity_1
3468	1995-01-08	EL	5	1995	1	0.616448	17.390
3361	1995-01-08	EL	4	1995	1	0.558864	21.079
5036	1995-01-08	EL	3	1995	1	0.602376	8.920
3113	1995-01-08	EL	1	1995	1	0.621467	6.003
7739	1995-01-08	EL	2	1995	1	0.630947	34.043

```
In [10]: dat.tail()
```

```
Out[10]:
```

	date	area_type	area_number	iso_year	iso_week	filling_level	capacity_1
8263	2026-09-06	EL	4	2026	36	0.902762	21.07
14875	2026-09-06	VASS	1	2026	36	0.570397	36.04
14873	2026-09-06	VASS	3	2026	36	0.826252	28.15
8264	2026-09-06	EL	2	2026	36	0.460863	34.04
14876	2026-09-06	VASS	2	2026	36	0.516978	23.23

```
In [11]: (dat["next_publication_date"] == "0001-01-01T00:00:00").sum()
```

Out[11]: np.int64(11322)

```
In [12]: placeholder = dat[
           dat["next_publication_date"] == "0001-01-01T00:00:00"
         ]

# Count placeholder dates per year
placeholder_counts = (
    placeholder.groupby("iso_year")
    .size()
    .reset_index(name="count")
)
placeholder_counts
```

Out[12]:

	iso_year	count
0	1995	468
1	1996	468
2	1997	468
3	1998	477
4	1999	468
5	2000	468
6	2001	468
7	2002	468
8	2003	468
9	2004	477
10	2005	468
11	2006	468
12	2007	468
13	2008	468
14	2009	477
15	2010	468
16	2011	468
17	2012	468
18	2013	468
19	2014	468
20	2015	477
21	2016	468
22	2017	468
23	2018	468
24	2019	54

The value "0001-01-01T00:00:00" in column "next_publication_date" does not represent a publication date, but seems to be a placeholder for missing publication dates. I therefore convert these values to NaT (not a time), which is pandas' representation of missing datetime values.

```
In [13]: # Replace placeholder dates with missing values
dat["next_publication_date"] = dat["next_publication_date"].replace(
    "0001-01-01T00:00:00",
    pd.NA
)
```

```
In [14]: dat.tail()
```

```
Out[14]:
```

	date	area_type	area_number	iso_year	iso_week	filling_level	capacity_
8263	2026-09-06	EL	4	2026	36	0.902762	21.07
14875	2026-09-06	VASS	1	2026	36	0.570397	36.04
14873	2026-09-06	VASS	3	2026	36	0.826252	28.15
8264	2026-09-06	EL	2	2026	36	0.460863	34.04
14876	2026-09-06	VASS	2	2026	36	0.516978	23.23

Exploring number of observations for each area number and each area type, and combinations of these.

```
In [15]: # distribution over areas
dat["area_number"].value_counts()
```

```
Out[15]: area_number
3      3306
1      3306
2      3306
5      1653
4      1653
0      1653
Name: count, dtype: int64
```

```
In [16]: # distributions across area types
dat["area_type"].value_counts()
```

```
Out[16]: area_type
EL      8265
VASS    4959
NO      1653
Name: count, dtype: int64
```

```
In [17]: # distributions across area types and area numbers
dat[["area_type", "area_number"]].drop_duplicates().sort_values(
    ["area_type", "area_number"]
)
```

```
Out[17]:
```

	area_type	area_number
3113	EL	1
7739	EL	2
5036	EL	3
3361	EL	4
3468	EL	5
8501	NO	0
14162	VASS	1
11380	VASS	2
11198	VASS	3

Visualizations

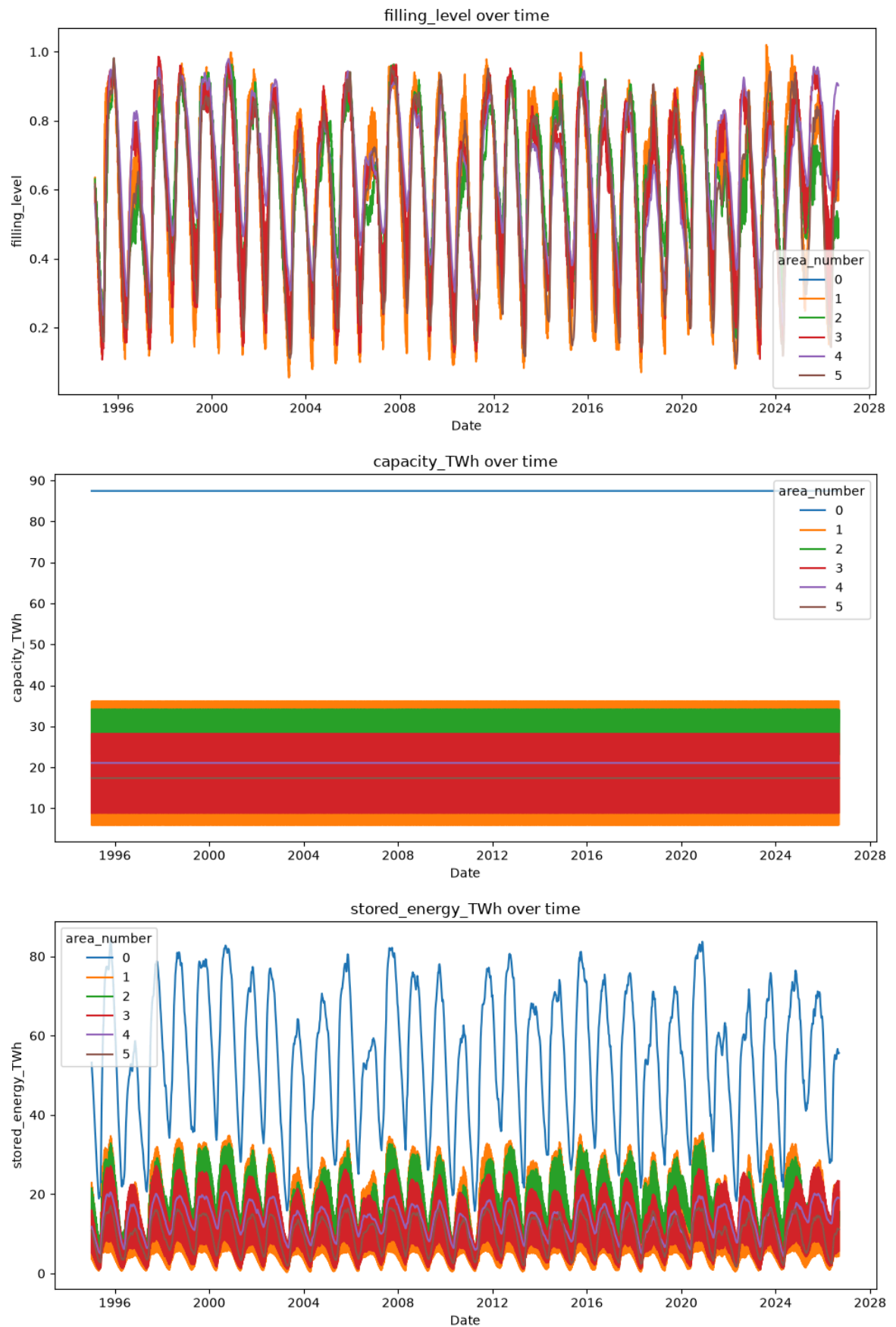
Plot of each column

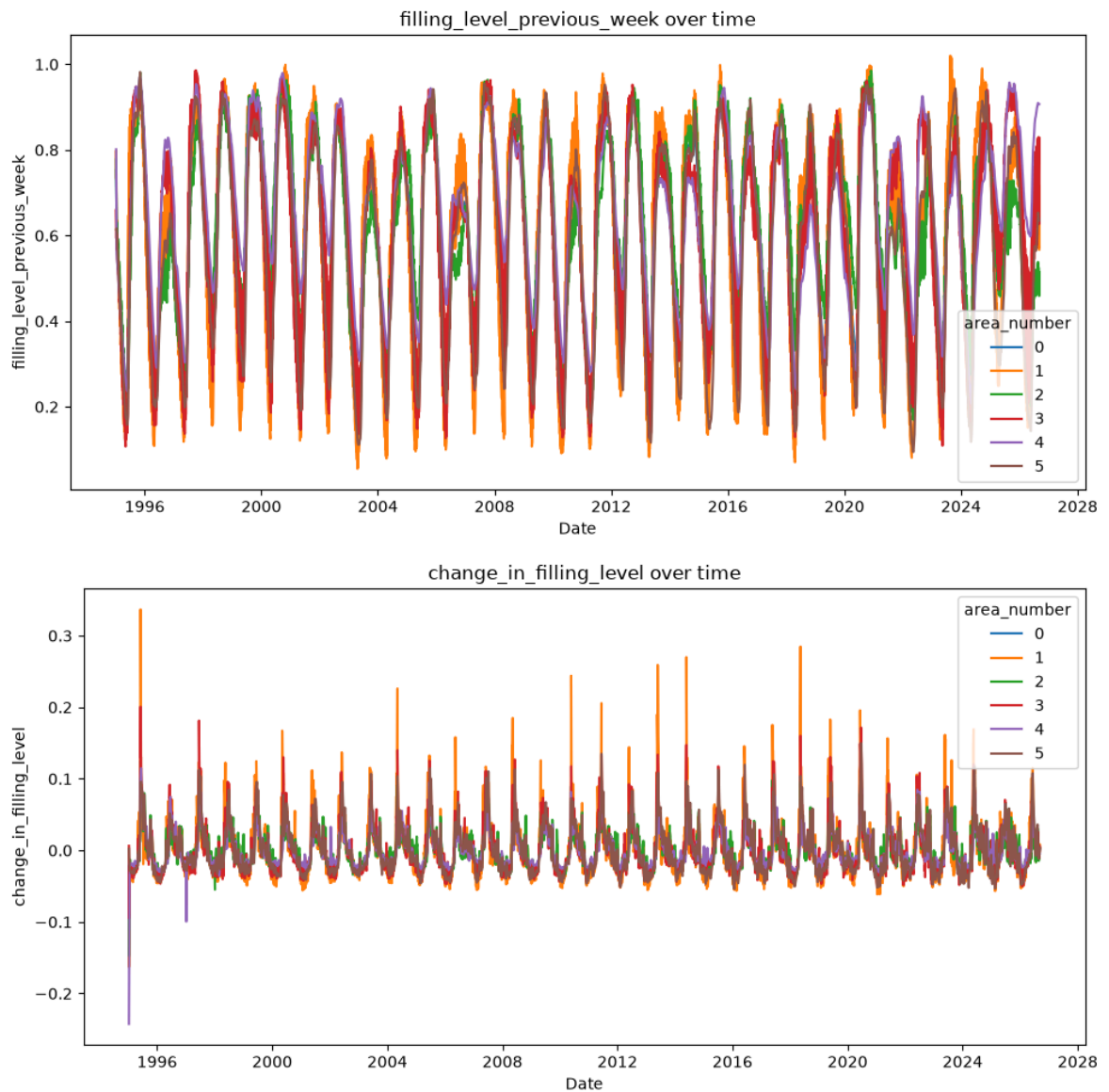
```
In [18]: # numerical measurement columns
line_cols = [
    "filling_level",
    "capacity_TWh",
    "stored_energy_TWh",
    "filling_level_previous_week",
    "change_in_filling_level"
]

# line plots
for col in line_cols:
    plt.figure(figsize=(10, 5))

    # Plot one line for each area number
    for area_number, group in dat.groupby("area_number"):
        plt.plot(
            group["date"],
            group[col],
            label=area_number
        )

    plt.title(f"{col} over time")
    plt.xlabel("Date")
    plt.ylabel(col)
    plt.legend(title="area_number")
    plt.tight_layout()
    plt.show()
```





Distributions of categorical and discrete variables

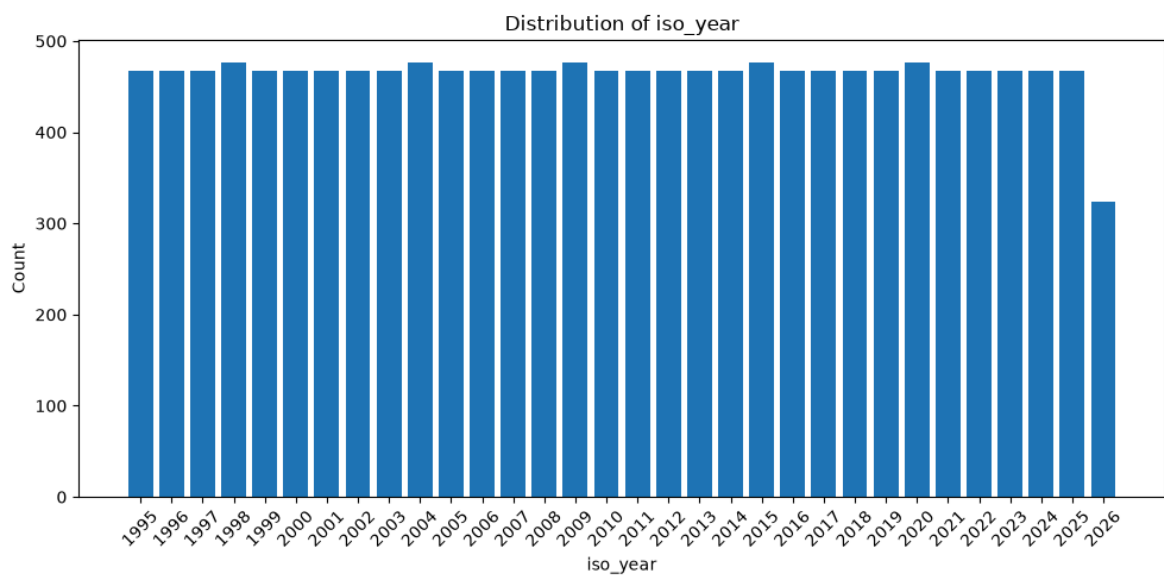
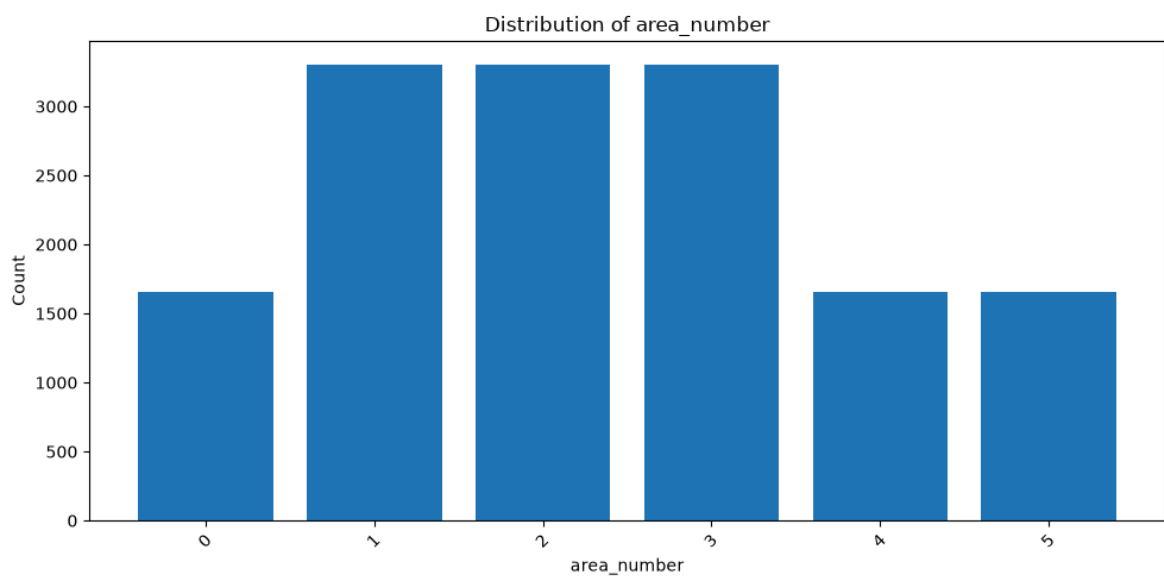
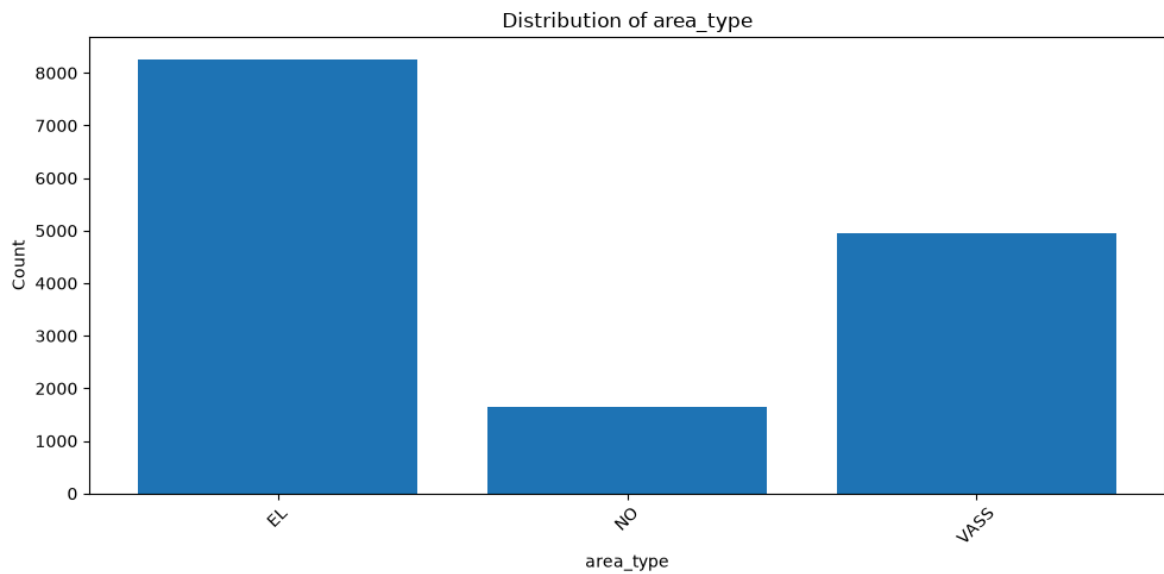
```
In [19]: categorical_cols = [
    "area_type",
    "area_number",
    "iso_year",
    "iso_week"
]

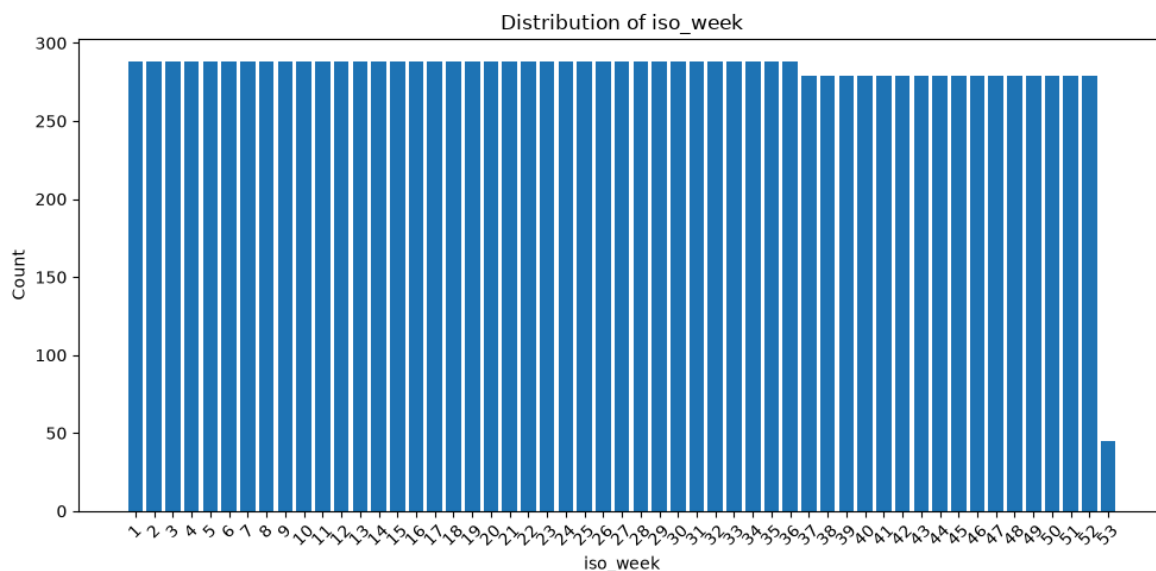
for col in categorical_cols:
    # Count observations for each variable
    count_data = (
        dat[col]
        .value_counts()
        .sort_index()
    )

    # Plot variable versus counts
    plt.figure(figsize=(10, 5))

    plt.bar(
        count_data.index.astype(str),
        count_data.values
    )
```

```
plt.title(f"Distribution of {col}")
plt.xlabel(col)
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```





Plot of all columns together

The measurement variables are on different scales, so I scale them before plotting them together. I use min-max scaling to transform each variable to a range between 0 and 1. This makes it easier to compare the patterns of the variables in the same plot.

```
In [20]: measure_cols = [
    "filling_level",
    "capacity_TWh",
    "stored_energy_TWh",
    "filling_level_previous_week",
    "change_in_filling_level",
    ]

# multiple observations for each date
plot_data = dat.groupby("date")[measure_cols].mean().reset_index()
```

```
In [21]: # Min-max scale each measurement column
for col in measure_cols:
    plot_data[col] = (
        (plot_data[col] - plot_data[col].min())
        / (plot_data[col].max() - plot_data[col].min())
    )
```

```
In [22]: # line plot of all measurement columns
plt.figure(figsize=(10, 5))

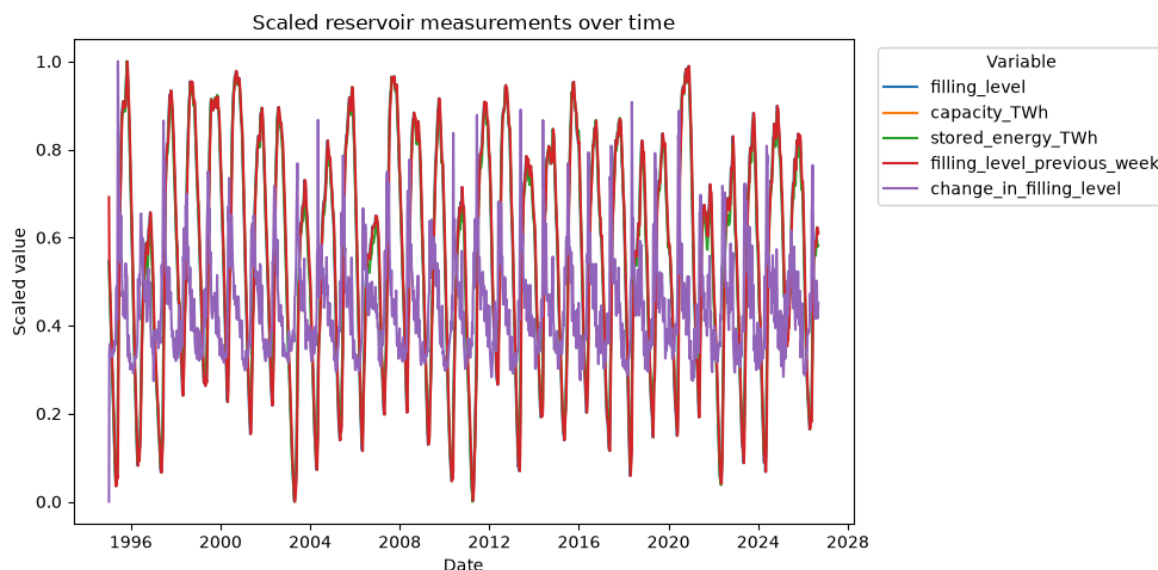
for col in measure_cols:
    plt.plot(
        plot_data["date"],
        plot_data[col],
        label=col
    )

plt.title("Scaled reservoir measurements over time")
plt.xlabel("Date")
plt.ylabel("Scaled value")
plt.legend()
```

```

title="Variable",
bbox_to_anchor=(1.02, 1),
loc="upper left")
plt.tight_layout()
plt.show()

```



Log (description of compulsory work)

Working with part one of the IND320 project has been both fun and challenging. I had no previous experience with Streamlit or making apps in general, so I have definitely learned a lot.

When working with the notebook, I was first unsure about what kind of data I was working with. After some inspection, googling, and prompting ChatGPT, I learned more about the data. It contains data about Norwegian reservoir levels and stored energy over time. The data contains different area types, including EL, VASS, and NO. The area numbers overlap between the different area types. EL includes area numbers 1 to 5, while VASS contains area numbers 1, 2, and 3. Understanding the data also helped me when translating the column names into meaningful English names.

My biggest challenge while working on the notebook was deciding how to plot the features. When plotting each feature separately, I decided to plot the numerical measurement columns using line plots and use bar plots for the remaining categorical variables. For the combined plot, I decided to only plot the numerical measurement columns. I did not think it made sense to include the features related to time, such as "date", "iso_year", and "iso_week". Since there are several observations for each date, I also had to consider how to handle these observations. I did not do anything about this when plotting the columns separately, but when plotting the measurement columns together, I grouped the data by date and calculated the mean across the different areas. This makes the plot easier to interpret, since I am interested in the overall variation in the variables and differences between the variables rather than the variation for each individual area. I also used min-max scaling before plotting the measurement variables together because they

are measured on different scales. When plotting the columns separately, I color the measurement columns according to area number to distinguish between the different area numbers.

When working with the Streamlit app, I used the IND320_student repository as inspiration for how to structure the folders and files. I had no previous experience with Streamlit, so I looked at the lecture notes and completed the class exercises to understand how it works. I also needed some help from ChatGPT to implement features such as `st.select_slider` and `LineChartColumn`, in addition to using Google to find information. Making the different pages took some trial and error.

My main challenge was again deciding which columns to include when plotting line charts and which features to include when plotting all columns. I did not think it made sense to plot "iso_year", "iso_week", and "date" as line plots, so I again decided to only plot the measurement columns. On the plot page (page 3), I also decided to plot the mean of the observations for each date when plotting all measurement columns together. When a single variable is selected, I use `area_number` to distinguish between the different areas in the plot.

Working with Git has gone smoothly. I created a new branch for this assignment and merged it into main when I was finished.

Description of AI usage

Prompt to chatGPT: Can you write a short summary about how I have used you in this assignment?

I used ChatGPT as a support tool throughout the project. It was mainly used to help me understand the assignment requirements, translate the original Norwegian variable names into understandable English names, troubleshoot errors, and discuss appropriate ways to visualize the reservoir data. I also used ChatGPT for guidance when developing the Streamlit app, including the page structure, data filtering, interactive widgets, and deployment. The final code was implemented and tested by me, and I used the explanations to better understand the choices made throughout the project.

GitHub Copilot was disabled while I was working on this assignment.